# The Effects of Monetary Policy in Colombia Revisited: Policy Shocks vs. Information Shocks


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**Abstract**

This paper estimates the effect of monetary policy on output and inflation in Colombia distinguishing between pure monetary policy shocks and information shocks embedded in high-frequency policy surprises following Jarociński and Karadi (2020). To do this, the paper exploits high-frequency information on the comovement between break-even, short-term, market inflation expectations (BEI) and overnight index swaps (OIS) on the short-term interest rate around policy announcements by the Banco de la República. Policy shocks are characterized by a negative comovement between those variables around policy announcements, whereas information shocks will be characterized by the opposite comovement. Using these shocks, the paper deploys the local projections method (Jordà (2005)) to estimate the dynamic responses of output and inflation to them. The response of the price level to a monetary policy surprise exhibits a "price puzzle", which largely disappears when estimating the response of the price level to pure policy shocks. The response of output is consistently negative under both scenarios, although volatile and less precisely estimated. Finally, the price level increases strongly in response to information shocks.



**Keywords:** Monetary Policy Surprises, Monetary Policy Shocks, Information Shocks, Break-Even Inflation.
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What are the effects of monetary policy on the macroeconomy? This paper revisits this question for the case of Colombia distinguishing between pure monetary policy shocks and information shocks embedded in high-frequency policy surprises following Jarociński and Karadi (2020).

As has been long established[1], finding the answer to the longstanding question of the effects of monetary policy relies critically on the ability of researchers to isolate monetary policy shocks, that is, variations in the policy instrument that are plausibly not a response to the past, current, or future state of the economy itself. In principle, these shocks can be used to calculate the impact of pure policy variations on the economy. Over the years, different methods have been proposed to isolate these shocks. Among them, one that has become popular in recent times is to exploit the richness of high-frequency financial markets information (HFI) close and around those snapshots of time when Central Banks make announcements on policy decisions.

Assuming that the state of the economy is observed to the same extent by both the Central Bank and market participants, and that there are no differences between the forecasts of market participants and the Central Bank given the same information set, all announced policy decisions that differ from the expectation of market participants very shortly before the announcements are plausible policy shocks, and all changes in financial markets prices around the same time window are a pure consequence of them. In the canonical papers by Gertler and Karadi (2015) and Nakamura and Steinsson (2018), these changes are referred to as "policy surprises", which are supposed to isolate purely exogenous policy perturbations (while the response of the Central Bank to the state of the economy is already captured by prices and by the expectations of market participants right before the announcements)[2].

Despite the increasing popularity of HFI methods, the necessary assumptions for surprises to serve as policy shocks are rarely met in practice. At least since Romer and Romer (2000), it has been widely accepted that the information set of the Central Bank with regards to the economy is different than that of market participants. Therefore, policy surprises include pure policy shocks combined with an "information revealing" effect. At its core, this is what Ramey (2016) refers to as the "foresight problem":

[1] See the chapter review by Ramey (2016).

[2] In the case of Colombia, Romero et al. (2021) are an example of the use of this technique to study monetary policy transmission. See also Kohlscheen (2014) for an example of the use of high-frequency changes in exchange rates around policy announcements to study the issue of the "exchange rate puzzle".

when the Central Bank reacts to its own expectations with regards to the state of the economy, and market participants have a different information set than that of the Central Bank, policy surprises are the sum of the pure policy shock and "news" about the expectation of the Central Bank with regards to key macroeconomic aggregates[3].

An example might help to reveal this problem. In January 2026, the Board of Directors of the Banco de la República (the Central Bank of Colombia) decided to increase its policy rate by 100 basis points (bps) as a response to inflationary concerns arising from a large Government-decreed increase in the nationwide minimum wage. Before the meeting of the Board, market analysts were, by and large, expecting a milder increase of 50 bps. After the policy announcement, if the policy surprise were a pure policy shock, inflation expectations should have fallen as an immediate response of the economy to a suddenly tighter stance of monetary policy. Instead, short-term inflation expectations (measured as the break-even between nominal and CPI-indexed government debt securities) increased significantly. This illustrates that the policy change, together with a tightening of the monetary policy stance, revealed to participants in financial markets news about the Central Bank's expectation for short-term inflation, which they duly incorporated in their forecasts.

Therefore, in order to estimate the effects of monetary policy on the macroeconomy, it is necessary to separate pure monetary policy shocks and news or "information" shocks embedded in policy surprises. To achieve this, this paper builds from Jarociński and Karadi (2020) by exploiting HFI on the comovement between break-even, short-term, market inflation expectations (BEI) and overnight index swaps (OIS) on the short-term interest rate around policy announcements by the Banco de la República. According to a standard, New Keynesian model of the economy (see Galí (2015)), a persistent contractionary policy shock reduces inflation expectations. In contrast, a purely information shock increases inflation expectations. Following from this, policy shocks will be characterized by a negative comovement between OIS changes and BEI changes around policy announcements, whereas information shocks will be characterized by the opposite comovement. Once isolated the pure policy shocks, the paper deploys the local projections method (Jordà (2005)) to estimate the dynamic response of real activity and the price level to those shocks.

The main findings of the paper can be summarized as follows. As has been

[3]Cieslak and Schrimpf (2019) estimate that these news are larger than the purely shock component in about 40% of policy announcements.

observed in the literature, the response of the price level to a monetary policy surprise exhibits a “price puzzle” whereby a positive surprise induces an increase in the price level in the short term. This indicates that the response of the price level embeds the conventional contractionary effects of monetary policy with an additional informational component that induces an initial increase in prices. When estimating the response of the price level to pure policy shocks, the price puzzle disappears, according to standard theoretical predictions. The response of output is consistently negative under both scenarios, although volatile and less precisely estimated. Consistent with these results, the price level increases strongly in response to information shocks, an intuition already explained above. In this case, output exhibits a weaker response to the information shock.

Related literature, three strands: the main one is the effect of mp shocks on the economy (Ramey). El segundo es HFI shocks en general y más especificamente en pirozhkova2024trouble,checo2024monetary, bolhuis2024new , y el tercero es informational differences (Romer y Romer es el origen y pasando por Nakamura all the way to Jarocinsky). La contribucion es explotar directamente expectativas de inflacion proveniente de los mercados para construir shocks (intuicion de Cochrane en Ramey). Similar to Jarocinsky, we use market info, but unlike them we use inflation expectations, as stock market prices depend on oil prices in Colombia.

This paper unfolds as follows. Section 1 presents the two steps of the empirical strategy of this paper: the identification of policy and information shocks and the estimation of the effect of monetary policy on the macroeconomy. Section 2 briefly presents the data employed in the paper, while Section 3 presents the main results of the estimations. Some final thoughts are presented at the end as concluding comments.

# 1 Methodology

As explained before, this paper follows a two-step methodology. The first step consists in isolating policy shocks and information shocks using HFI on the comovement between break-even, short-term, market inflation expectations (BEI) and overnight index swaps (OIS) on the short-term interest rate around policy announcements by the Banco de la República. This step follows closely Cieslak and Schrimpf (2019) and Jarociński and Karadi (2020). The second step is the estimation of the dynamic response of output and inflation to these shocks using a standard local projections (LP) technique (Jordà (2005)).

## 1.1 Policy Surprises, Policy Shocks and Information Shocks

This paper measures monetary policy surprises using Overnight Index Swaps (OIS) on the IBR (*Indicador Bancario de Referencia*), the key market reference for a credit risk-free overnight rate in Colombia. These contracts involve the exchange of a fixed interest rate for a floating rate determined by the compounded overnight IBR over a given maturity. As a result, the fixed rate embedded in the contract summarizes market expectations about the future path of short-term interest rates at the time of pricing. This paper uses the OIS-implied rate for the following month, observed during the same week as the meeting of the Board of Governors of Banco de la República, as the market expectation of the short-term future policy rate $i_t^M$ shortly before a policy announcement at time $t$, $E[i_t^M \mid t-\delta]$. Here, $t-\delta$ denotes the information set available to market participants shortly before the policy announcement.

In developed economies, HFI identification of pure policy shocks typically relies on narrow windows around monetary policy announcements (Cesa-Bianchi et al., 2020; Gertler & Karadi, 2015). In these settings, $\delta$ is often defined as a 30-minute window around the announcement (Nakamura and Steinsson (2018)). However, this approach is generally not feasible in emerging economies, where financial markets are often closed when the policy decision is announced (Kohlscheen, 2014). In the case of the Colombian OIS market considered here, the relevant information becomes available on the day the OIS-implied rate for the following month is revealed, which occurs a few days before the monetary policy meeting.

Monetary policy surprises are then defined as the difference between the realized policy rate and the market-implied expected rate:

$$MPS_t = i_t^M - E[i_t^M \mid t-\delta].$$

In this specification, $MPS_t$ captures the unanticipated component of the monetary policy announcement.

As explained above, $MPS_t$ is likely to embed, beyond exogenous policy innovations, news or information shocks with regard to the future state of the economy. The latter might reflect, for instance, an informational or technological advantage held by the Central Bank in the construction of relevant forecasts, and is partially revealed when the policy decision is announced (Bauer & Swanson, 2023; Miranda-Agrippino, 2016; Nakamura & Steinsson, 2018). To isolate these two components, this paper decomposes

$MPS_t$ into a pure monetary policy shock and a Central Bank information shock, using an identification strategy closely related to Cieslak and Schrimpf (2019) and Jarociński and Karadi (2020). While those papers exploit the joint high-frequency response of interest rates and stock prices around policy announcements, this paper instead uses the joint response of monetary policy surprises and short-term, market, break-even, inflation expectations. The underlying logic is that high-frequency financial-market reactions reveal whether the announcement is interpreted primarily as a policy innovation or as news about fundamentals. (Jarociński & Karadi, 2020) :contentReference[oaicite:2]index=2

More specifically, if $MPS_t$ captured only a pure monetary policy shock, an unexpected tightening should be followed by a decline in market inflation expectations. Such a response would reflect the standard transmission mechanism of a standard, New Keynesian, model: persistently tighter policy reduces expected demand and expected inflationary pressures. Conversely, if inflation expectations increase after the announcement, the surprise cannot be understood as purely monetary. Rather, it indicates that the central bank has revealed information consistent with stronger inflationary pressures than those previously priced by the market. In that case, the observed surprise reflects plausibly both a contractionary policy innovation and an information effect working in the opposite direction on expectations. This observation provides the basis for this decomposition: the high-frequency comovement of $MPS_t$ and market inflation expectations allows the isolation of pure monetary policy shocks and central bank information shocks.

Market inflation expectations are measured using breakeven inflation (BEI), defined as the difference between nominal TES bond yields and real UVR-indexed bond yields at a given horizon, $h$. BEI is a key market-based indicator for assessing the degree of anchoring of long-term inflation expectations. The change in BEI around the announcement, $\Delta BEI_t$, provides the high-frequency change in inflation expectations used in the methodology described below.

$$\Delta BEI_t = E[\pi_{t+h} \mid t+\delta] - E[\pi_{t+h} \mid t-\delta].$$

In the empirical implementation, $t+\delta$ corresponds to the business day immediately after the announcement, while $t-\delta$ corresponds to the business day immediately before the announcement, and $h$ is set at 1 year.[4] Figure

[4]BEI may in principle contain information other than inflation expectations, such as

1 presents a scatterplot of $MPS_t$ and $\Delta BEI_t$. The figure shows that, for some policy announcements, unexpected positive monetary surprises are accompanied by increases in market-based inflation expectations. These observations are interpreted as evidence that the monetary policy surprise contains a relevant central bank information component.

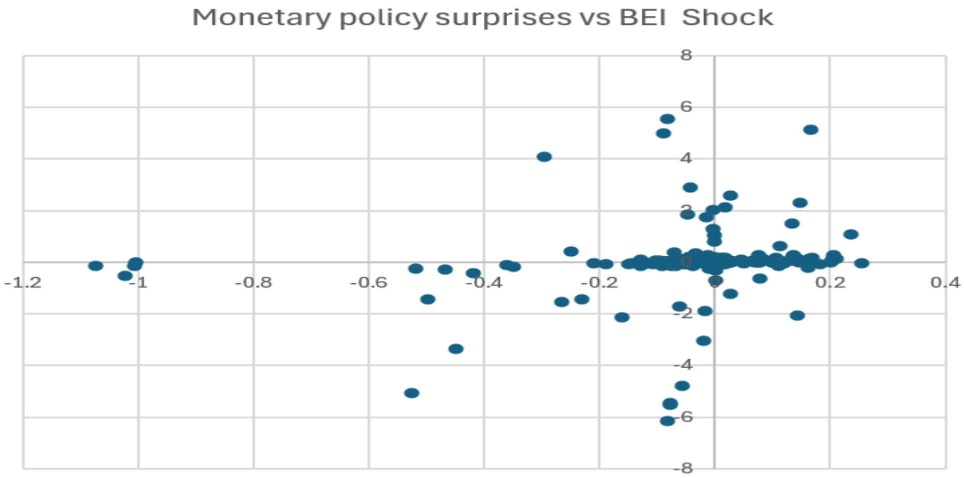


Figure 1: Scatter plot of high frequency movements in BEI and Monetary Policy Surprises.

Note: Scatter plot of $MPS_t$ and $\Delta BEI_t$ around monetary policy announcements.

Using these two high-frequency measures, this paper implements the procedure proposed by Jarociński and Karadi (2020). This procedure generates two orthogonal shocks: one associated with the central bank information component, $u_t^i$, and another associated with pure monetary policy shocks, $u_t^m$. Identification relies on the information embedded in $\Delta BEI_t$ and follows the sign restrictions of Table 1. Specifically, $u_t^i$ is expected to be active whenever there is a positive comovement between $\Delta BEI_t$ and $MPS_t$, whereas $u_t^m$ is expected to be active when the comovement is negative. A complete description of the methodology is presented in Appendix A.

## 1.2 The Effects of Shocks

To estimate the dynamic effects of both policy shocks and information shocks on the consumer price index and GDP, this paper deploys a battery of local projections, following Jordà (2005). For each horizon $h = 0, 1, \ldots, H$, the paper estimates the following econometric specification:

$$P_{t+h} = \alpha_h + \beta_h S_t + \gamma_h P_{t-1} + X_t \Gamma_h + \varepsilon_{t,h}, \tag{1}$$

inflation risk premia and liquidity premia (Espinosa-Torres et al., 2017) which are not cleaned up in this paper.

| Variable | $u_t^m$ | $u_t^i$ |
|---|---|---|
| $MPS_t$ | + | + |
| $\Delta BEI_t$ | - | + |

Table 1: Sign restrictions for identification.

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Note: Sign restrictions to identify central bank information componetn ($u_t^i$) and raw monetary policy component ($u_t^m$).

where $P_{t+h}$ denotes the variable of interest (output or the price level) at horizon $h$, and $S_t$ is one of the shocks identified in the previous section, namely the monetary policy surprise $MPS_t$, the pure monetary policy shock $u_t^m$, or the central bank information shock $u_t^i$. The coefficient $\beta_h$ captures the dynamic effect of the corresponding shock on variBLE $P$ at horizon $h$.

Following Bauer and Swanson (2023), even high-frequency monetary policy surprises may remain correlated with macroeconomic conditions observed prior to the policy announcement. To address this concern, the local projection specification includes the vector of observables $X_t$, which contains lagged macroeconomic controls, such as GDP, CPI, and the interest rate, that summarize the information available before the shock is realized. These controls help isolate the exogenous component of $S_t$. The paper estimates the specification separately for each shock using OLS at each horizon and compute inference using bootstrap standard errors.

# 2 Data

The relevant outcomes of the analysis are the logarithm of the consumer price index and the logarithm of GDP. The CPI corresponds to the monthly consumer price index published by the National Administrative Department of Statistics in Colombia (*DANE*, for its Spanish acronym). Since GDP is only available at quarterly frequency, we construct a monthly GDP series by temporally disaggregating the official quarterly real GDP using the monthly economic activity indicator (*ISE*) produced by *DANE* as an auxiliary indicator. To do so, we use the procedure in Sax and Steiner (2013), following the methodology of Chow and Lin (1971). In addition, we include the monthly average of interbank interest rate, which closely tracks the behavior of the monetary policy rate.

Inflation expectations and policy surprises are constructed as explained

above and relevant information is sourced from the Banco de la República's Market Analysis Department.
Data are monthly and span the period from 2008:1 to 2019:12. The sample starts in 2008 because surprises using the OIS market are only available from that date onward.[5]

# 3 Results

## 3.1 Baseline Results

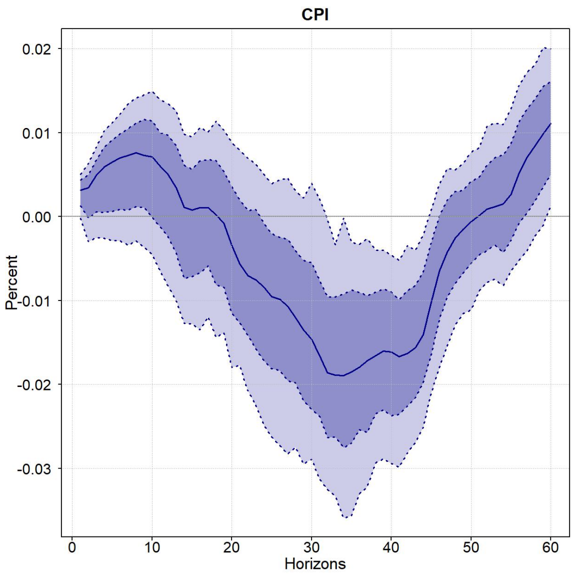
In this subsection, we estimate equation 1 using the three alternative measures of monetary policy shocks, namely $MPS_t$, $u_t^r$, and $u_t^i$. Figure 2 reports the responses of CPI and GDP when the shock is measured by the monetary policy surprise, $MPS_t$. The response of CPI displays a non-monotonic pattern. In the short run, prices increase and reach a local maximum of about 0.8% after roughly 8 to 10 months. This initial increase is consistent with the price puzzle often found in the high-frequency identification literature. Afterward, the response turns negative, reaching its trough around month 32, with a decline close to 2% in the price level. GDP also declines on impact, although the response is substantially more volatile and less precisely estimated. Its largest contraction occurs within the first half of the horizon, and the point estimate suggests a fall of around 1.5% to 2.0%. Overall, Figure 2 suggests that the raw surprise $MPS_t$ combines the conventional contractionary effects of monetary policy with an additional component that generates an initial increase in prices.

Figure 3 reports the same exercise using $u_t^r$, the raw monetary policy component obtained from the decomposition. Two features stand out. First, the short-run increase in CPI observed in Figure 2 largely disappears. Instead, the response of prices becomes negative much earlier and reaches a trough of about 2.4% around month 33. This pattern is more in line with the standard prediction that a contractionary monetary policy shock lowers prices. Second, GDP also falls after the innovation, with the strongest decline taking place during the first part of the horizon. Relative to the response obtained with $MPS_t$, the contraction in output appears somewhat cleaner, although the confidence bands remain wide. Taken together, these results indicate that once the information component is removed from the raw surprise, the estimated effects of monetary policy become more consistent with conventional theory and the price puzzle is substantially

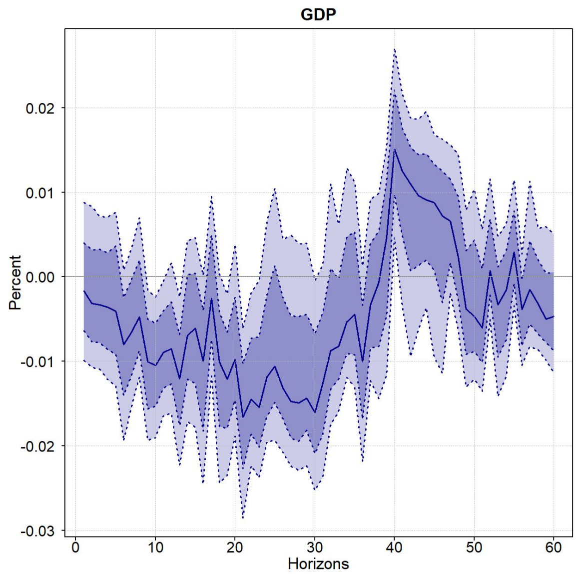
[5]We end the sample in 2019 in order to exclude the distortions associated with the COVID-19 period.

attenuated.

Finally, Figure 4 presents the responses to $u_t^i$, the central bank information component. The response of CPI is positive in the short run, peaking at around 1.1% to 1.2% during the first year, which suggests that the initial increase in prices observed in Figure 2 is closely related to this information channel. At longer horizons, however, the CPI response becomes negative before reverting again. Hence, the contribution of $u_t^i$ is most clearly associated with the short-run price increase, rather than with a uniformly positive response over the entire horizon. For GDP, the response is again imprecisely estimated, but the point estimates suggest that the information shock does not generate the same immediate contraction as the purified monetary policy shock. Overall, the comparison across figures supports the view that the raw monetary policy surprise $MPS_t$ conflates two different forces: a conventional contractionary monetary policy shock and a central bank information shock. Once the two are separated, the former is associated with lower prices and lower output, while the latter helps explain the short-run increase in prices that gives rise to the price puzzle.



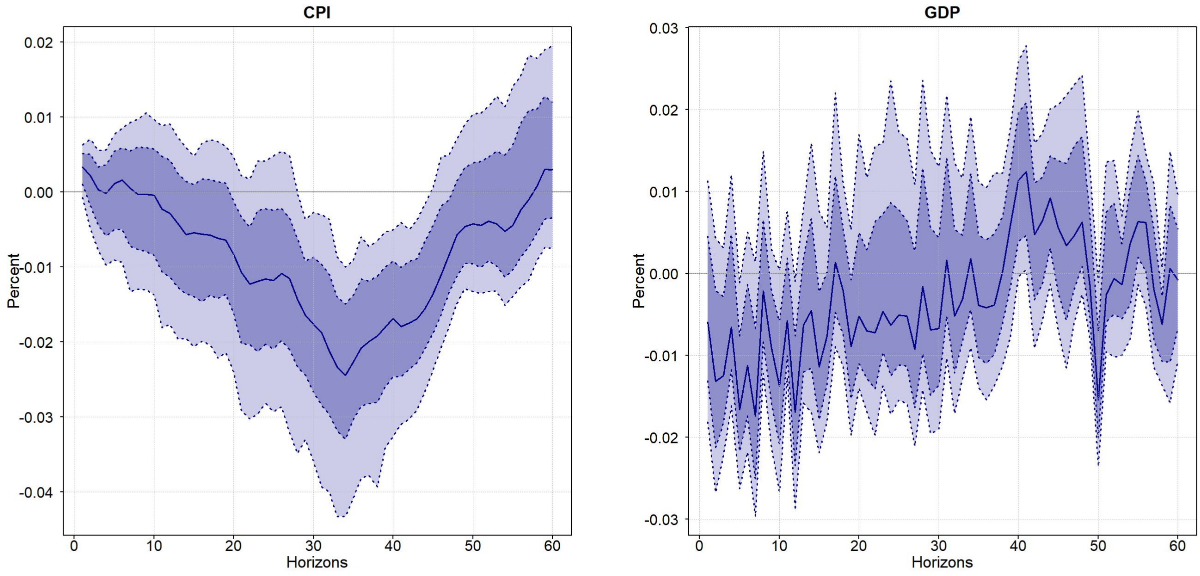

(a) IRF of CPI.




(b) IRF of GDP.

Figure 2: IRFs to an unexpected increase in the interest rate using the $MPS_t$ as shock.

Note: Panel (a) reports the impulse response of CPI to a monetary policy surprise, and panel (b) reports the impulse response of GDP to the same shock. Shaded areas are 68.0% and 90.0% confidence intervals.

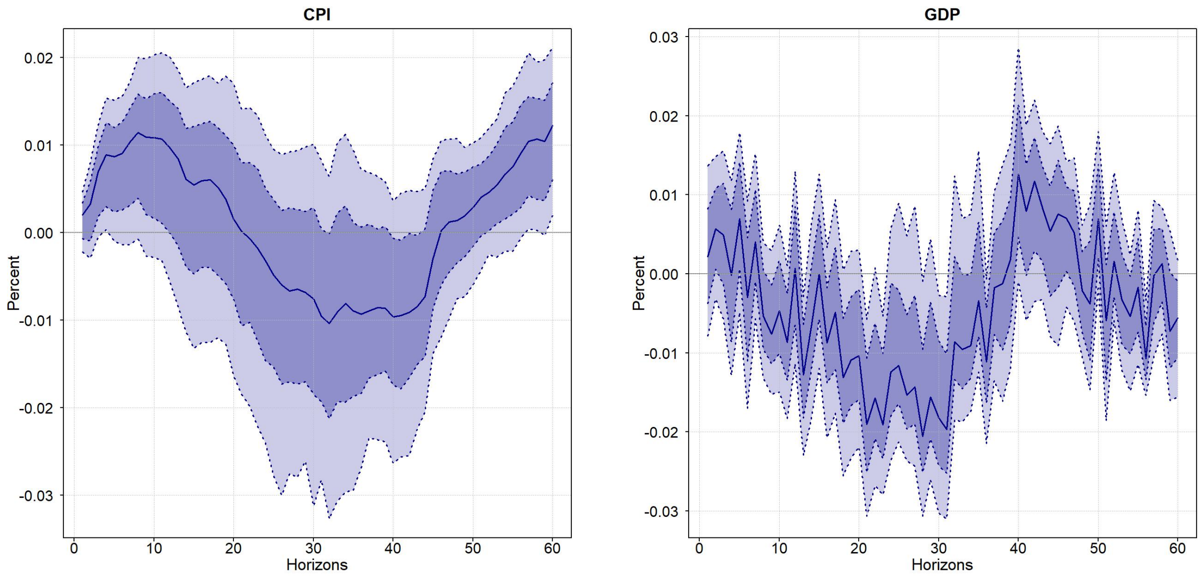



(a) IRF of CPI.

(b) IRF of GDP.

Figure 3: IRFs to an unexpected increase in the interest rate using the $u_t^r$ as shock.

Note: Panel (a) reports the impulse response of CPI to a raw monetary policy shock, and panel (b) reports the impulse response of GDP to the same shock. Shaded areas are 68.0% and 90.0% confidence intervals.




(a) IRF of CPI.

(b) IRF of GDP.

Figure 4: IRFs to an unexpected increase in the interest rate using the $u_t^i$.

Note: Panel (a) reports the impulse response of CPI to a central bank information component shock, and panel (b) reports the impulse response of GDP to the same shock. Shaded areas are 68.0% and 90.0% confidence intervals.

## 3.2 Robustness Checks

In addition to the baseline estimations, this paper performs two exercises: first, we check that the main conclusions of the results in the previous subsection do not depend on the exact measure of monetary policy surprises. We here rely on surprises coming from Bloomberg surveys as in Checo et al. (2024). Next, we use stock market prices to decompose the monetary policy surprises as in the document of Jarociński and Karadi (2020). We conclude that for Colombia, this alternative leads to puzzling behaviour, as it shows that the price puzzle is explained mainly by the raw monetary policy component. This could be related to the fact that stock market prices are closely correlated to exogenous variables (specifically, world oil prices) that make them unsuitable for properly identifying information shocks.

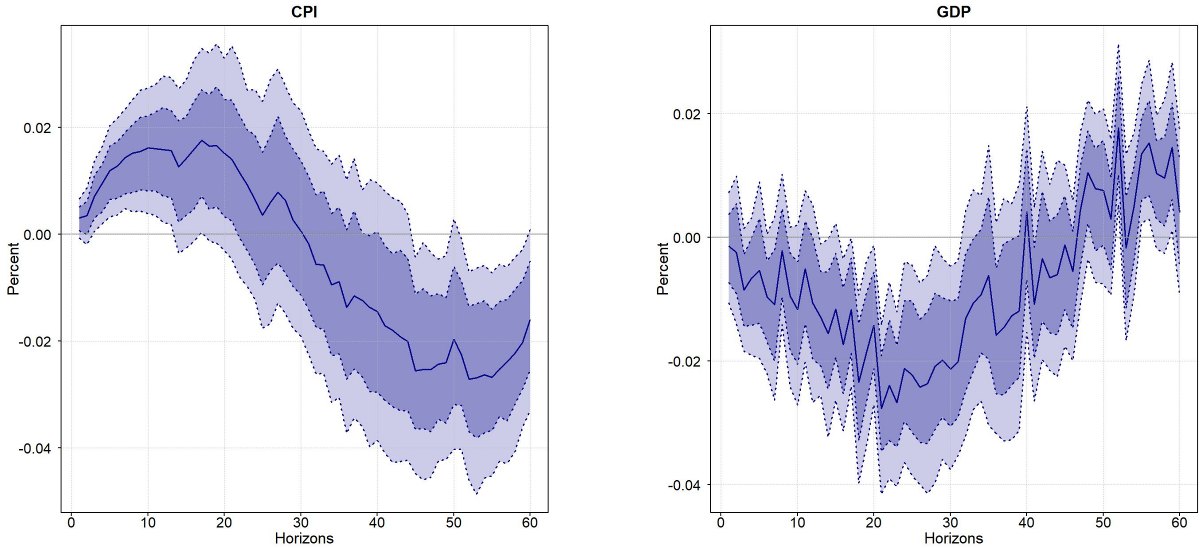
### 3.2.1 Alternative Monetary Policy Surprises

In this subsection, this paper follows the strategy of Checo et al. (2024), who build meeting-level forecast errors from Bloomberg policy-rate forecasts to mimic "high-frequency surprises", leveraging the fact that forecasts can be updated until very shortly before the meeting, so they should embed all the available information to market participants. Hence, the monetary policy surprise is constructed as the difference between the policy rate announced after the meeting and the average policy-rate forecast reported in Bloomberg surveys on the day of the meeting. Since these forecasts can be updated until the policy decision, they should incorporate the information available to market participants immediately before the announcement.

Figure 5 reports the responses of CPI and GDP when the monetary policy surprised is measured using the Bloomberg survey-based surprise. The results are very similar to those obtained with the OIS-based monetary policy surprise in Section 3. In particular, CPI increases in the short run after an unexpected tightening, displaying the same price-puzzle pattern documented in the baseline exercise. Afterward, the response turns negative, suggesting that the surprise again combines a conventional contractionary monetary policy effect with an additional component that pushes prices upward in the short run. The response of GDP is more volatile and less precisely estimated, but the point estimates indicate a contraction during the first part of the horizon. Overall, Figure 5 shows that the main features of the baseline results do not depend on the specific measure used to construct monetary policy surprises.

Figure 6 repeats the exercise using $u_t^r$, the raw monetary policy component obtained after decomposing the Bloomberg survey-based surprise. As in the baseline results, removing the central bank information component substantially attenuates the short-run increase in CPI observed when using the raw surprise. The response of prices becomes more negative over the horizon, which is consistent with the standard prediction that a contractionary monetary policy shock reduces inflationary pressures. The response of GDP also points to a contraction after the innovation, although the confidence bands remain wide. These results reinforce the interpretation that the short-run price puzzle is not mainly driven by the conventional monetary policy component. Instead, once the information component is separated from the raw Bloomberg surprise, the estimated effects of monetary policy are again more consistent with the usual transmission mechanism.

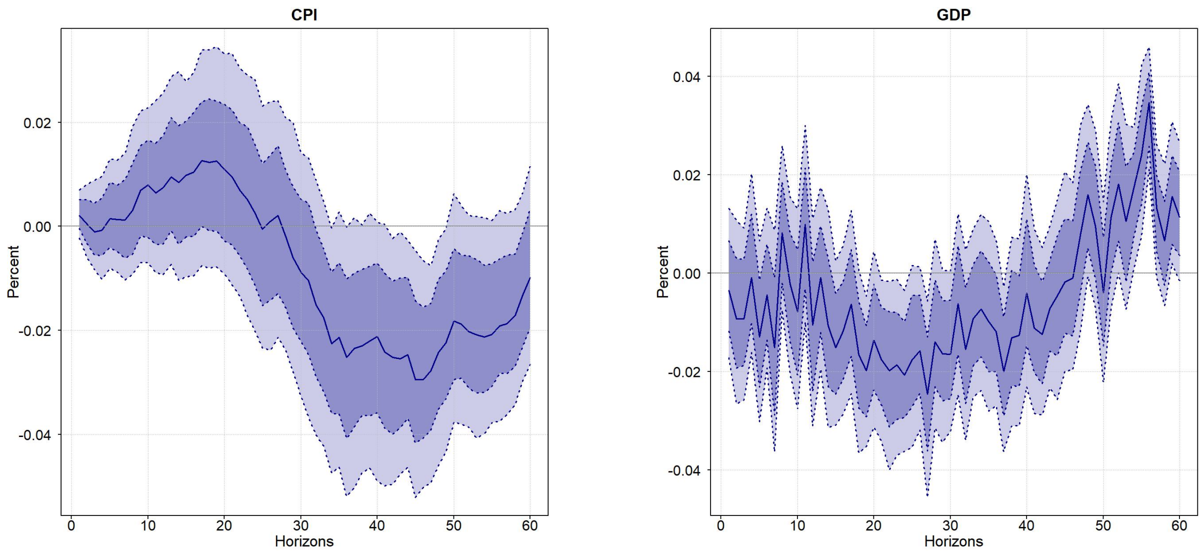
Finally, Figure 7 presents the responses to $u_t^i$, the central bank information component obtained from the Bloomberg survey-based surprise. The CPI response is positive in the short run, closely resembling the pattern found in the baseline specification. This result suggests that the initial increase in prices after a monetary policy surprise is largely associated with the information revealed by the central bank rather than with a pure contractionary policy shock. The GDP response is again imprecisely estimated, but it also becomes contractionary. Taken together, Figures 6–7 confirm that the main conclusion of Section 3 is robust to using an alternative measure of monetary policy surprises: the raw surprise conflates a conventional monetary policy shock with a central bank information shock, and the latter helps explain the short-run increase in prices that gives rise to the price puzzle.




(a) IRF of CPI. (b) IRF of GDP.

Figure 5: IRFs to an unexpected increase in the interest rate using $MPS_t$.

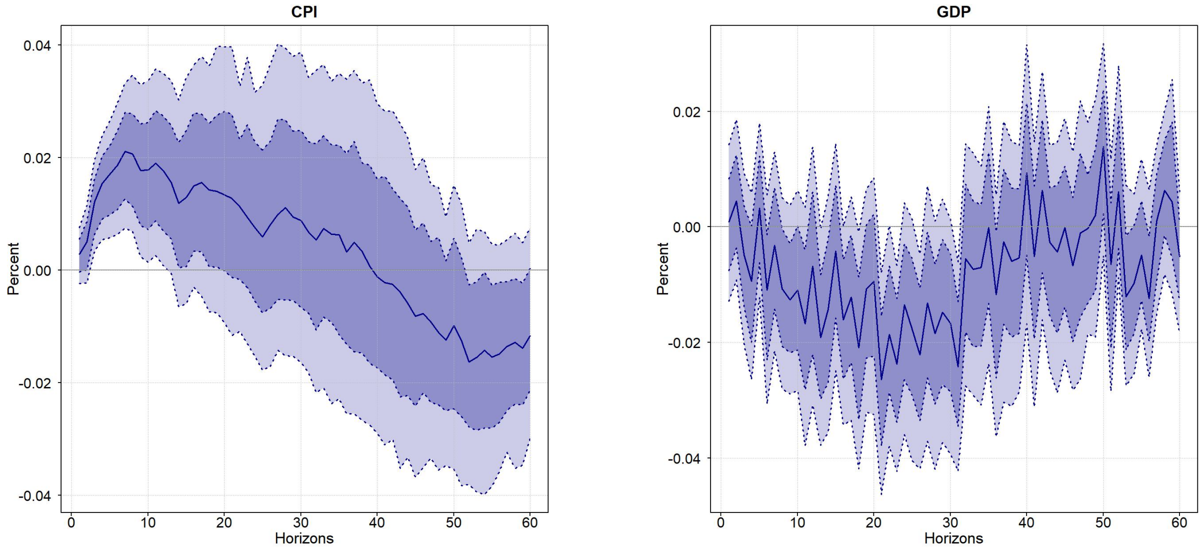
Note: $MPS_t$ is measured as the difference between the interest rate after the meeting and the expected interest rate in Bloomberg surveys. Panel (a) reports the impulse response of CPI to a monetary policy surprise, and panel (b) reports the impulse response of GDP to the same shock using the $MPS_t$. Shaded areas are 68.0% and 90.0% confidence intervals.




(a) IRF of CPI. (b) IRF of GDP.

Figure 6: IRFs to an unexpected increase in the interest rate using the $u_t^r$ as shock.

Note: When decomposing the surprises, $MPS_t$ is measured as the difference between the interest rate after the meeting and the expected interest rate in Bloomberg surveys. Panel (a) reports the impulse response of CPI to a raw monetary policy shock, and panel (b) reports the impulse response of GDP to the same shock. Shaded areas are 68.0% and 90.0% confidence intervals.




(a) IRF of CPI.

(b) IRF of GDP.

Figure 7: IRFs to an unexpected increase in the interest rate using the $u_t^i$.

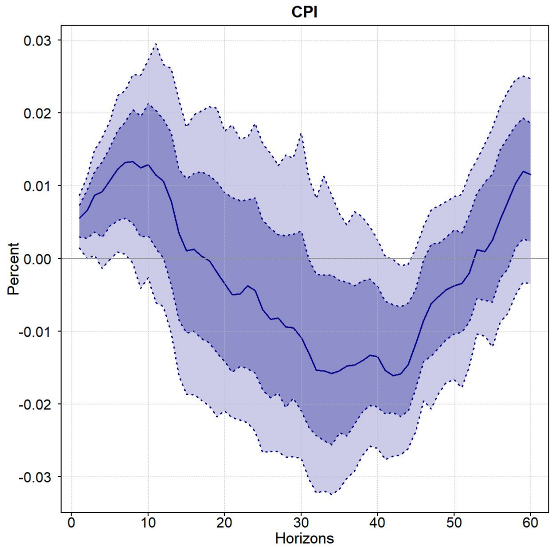
Note: When decomposing the surprises, $MPS_t$ is measured as the difference between the interest rate after the meeting and the expected interest rate in Bloomberg surveys. Panel (a) reports the impulse response of CPI to the central bank information component shock, and panel (b) reports the impulse response of GDP to the same shock. Shaded areas are 68.0% and 90.0% confidence intervals.

### 3.2.2 Stock Prices as an Alternative to BEI

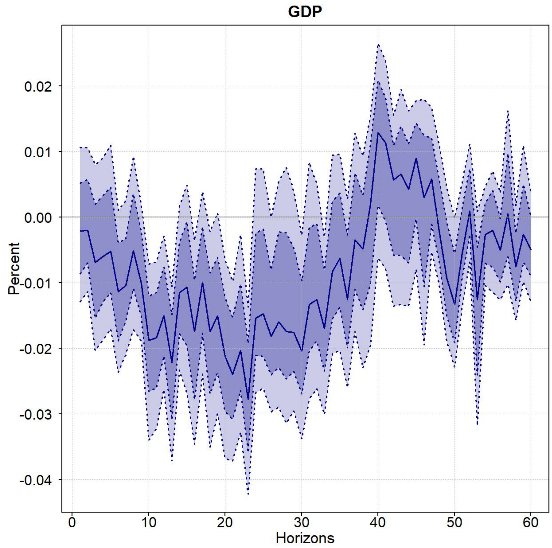
An alternative approach to identifying the central bank information component relies on the high-frequency response of stock market prices around monetary policy announcements. This strategy follows Jarociński and Karadi (2020), who argue that equity prices provide a useful signal to distinguish between conventional monetary policy shocks and information effects. The underlying intuition is straightforward. Under standard asset-pricing logic, an unexpected monetary tightening should lower stock market valuations, since higher interest rates raise discount rates and are typically associated with weaker expected future activity. By contrast, if stock prices increase following a positive monetary policy surprise, this suggests that the announcement may have also revealed favorable information about the economic outlook. In this subsection, we implement this alternative identification strategy for Colombia using the reaction of the COLCAP index, the benchmark stock market index for the Colombian economy. The response of the COLCAP around monetary policy announcements provides an alternative high-frequency signal to isolate the information content embedded in monetary policy surprises.

The decomposition based on the high-frequency response of the COLCAP leads to results that appear less consistent with the evidence obtained from break-even inflation. In the BEI-based exercise, the identifying variable is directly related to inflation expectations, so the information component captures the part of the monetary policy surprise associated with revisions in expected inflation. By contrast, the COLCAP-based decomposition relies on a different financial-market signal, reflecting channels that are not directly related to the inflation-expectations mechanism emphasized in this paper.

This distinction is reflected in the estimated responses. The COLCAP-based decomposition reallocates the price response across shocks in a way that differs from the BEI-based exercise and from the pattern emphasized by Jarociński and Karadi (2020). In the BEI-based decomposition, the central bank information component accounts for the initial increase in CPI, consistent with the interpretation that monetary policy announcements convey information about inflationary pressures. By contrast, when COLCAP is used as the financial-market signal, the price puzzle instead appears to be concentrated in the raw monetary policy component: CPI increases after the monetary policy shock during the first horizons before declining later (Figure 8).The CPI response to the information component is instead relatively muted at short horizons and becomes negative over the medium run (Figure 9). Hence, using COLCAP does not reproduce the same separation found in Jarociński and Karadi (2020), where a positive co-movement between interest rate surprises and stock prices is interpreted as favorable central bank information and is associated with higher real activity and prices. Rather, in the Colombian application, the stock-market-based decomposition appears to assign the puzzling short-run price increase mainly to the monetary policy component, suggesting that stock market valuations may provide a noisier signal than break-even inflation for identifying the inflation-expectations channel.



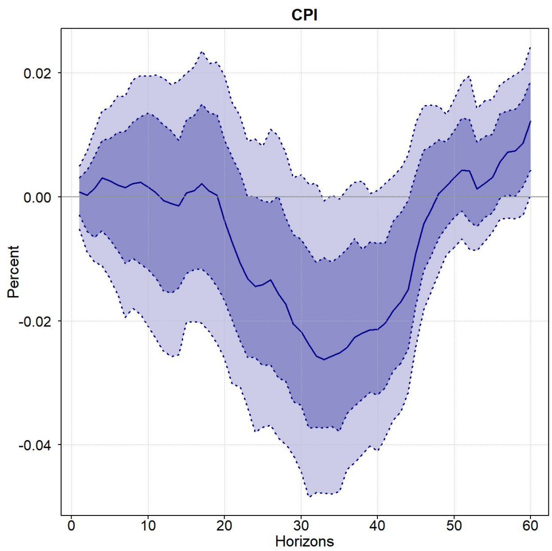

(a) IRF of CPI.



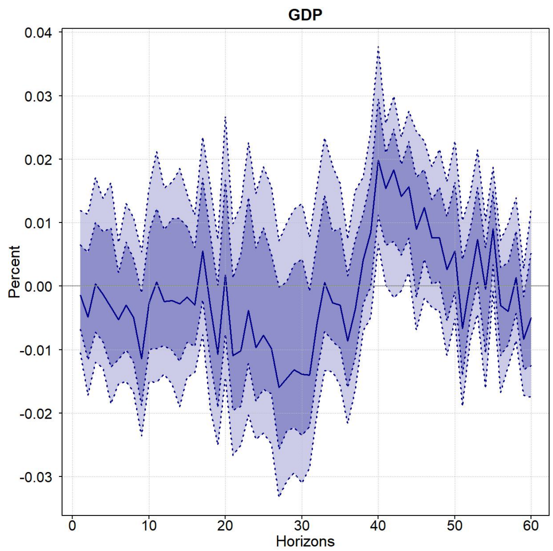

(b) IRF of GDP.

Figure 8: IRFs to an unexpected increase in the interest rate using the $u_t^r$ as shock.

Note: When decomposing the surprises, we use COLCAP changes to decompose the surprises. Panel (a) reports the impulse response of CPI to a raw monetary policy shock, and panel (b) reports the impulse response of GDP to the same shock. Shaded areas are 68.0% and 90.0% confidence intervals.




(a) IRF of CPI.




(b) IRF of GDP.

Figure 9: IRFs to an unexpected increase in the interest rate using the $u_t^i$.

Note: When decomposing the surprises, we use COLCAP changes to decompose the surprises. Panel (a) reports the impulse response of CPI to the central bank information omponent shock, and panel (b) reports the impulse response of GDP to the same shock. Shaded areas are 68.0% and 90.0% confidence intervals.

These results suggest that the COLCAP may not be the most appropriate financial variable to identify the central bank information component of monetary policy surprises in the Colombian case. One possible explanation is that movements in the COLCAP are strongly influenced by fluctuations in international oil prices. This is particularly relevant for Colombia because the stock market index has historically included a sizable participation of firms whose valuations are directly or indirectly exposed to the oil sector. As a result, changes in the COLCAP around monetary policy announcements may reflect not only the information revealed by the central bank, but also contemporaneous movements in global commodity prices that are unrelated to domestic monetary policy.

This concern is illustrated in Figure 10, which reports the annual percentage growth of the COLCAP and the West Texas Intermediate (WTI) oil price over the sample period. The two series display a strong positive association, with a correlation of 0.60, suggesting that Colombian stock market valuations are highly sensitive to oil price dynamics. This dependence is also present at higher frequencies. When we compute the change in WTI around monetary policy meetings, which should be exogenous to the information released by the Colombian central bank, its correlation with the high-frequency change in the COLCAP is 0.29. This value is economically relevant and indicates that part of the stock market reaction observed around policy announcements may be capturing global oil price movements rather than domestic monetary policy news. By contrast, the correlation between the high-frequency change in WTI and the high-frequency change in break-even inflation is only $-0.08$, suggesting that BEI is substantially less exposed to this external commodity-price factor.

This evidence provides a plausible explanation for why the COLCAP-based decomposition delivers a less clear separation between the monetary policy component and the central bank information component. If the stock market response partly reflects oil price movements, then the identifying signal used in the decomposition is contaminated by an external shock that is orthogonal to the monetary policy announcement itself. In that case, a positive co-movement between monetary policy surprises and the COLCAP cannot be interpreted exclusively as favorable central bank information about the domestic economic outlook. It may instead reflect the effect of oil price changes on the valuation of oil-related firms, on expected export revenues, or on broader perceptions about the Colombian economy as a commodity-exporting country. Therefore, the COLCAP-based signal may assign part of the observed price dynamics to the wrong structural component, generating misleading impulse responses.

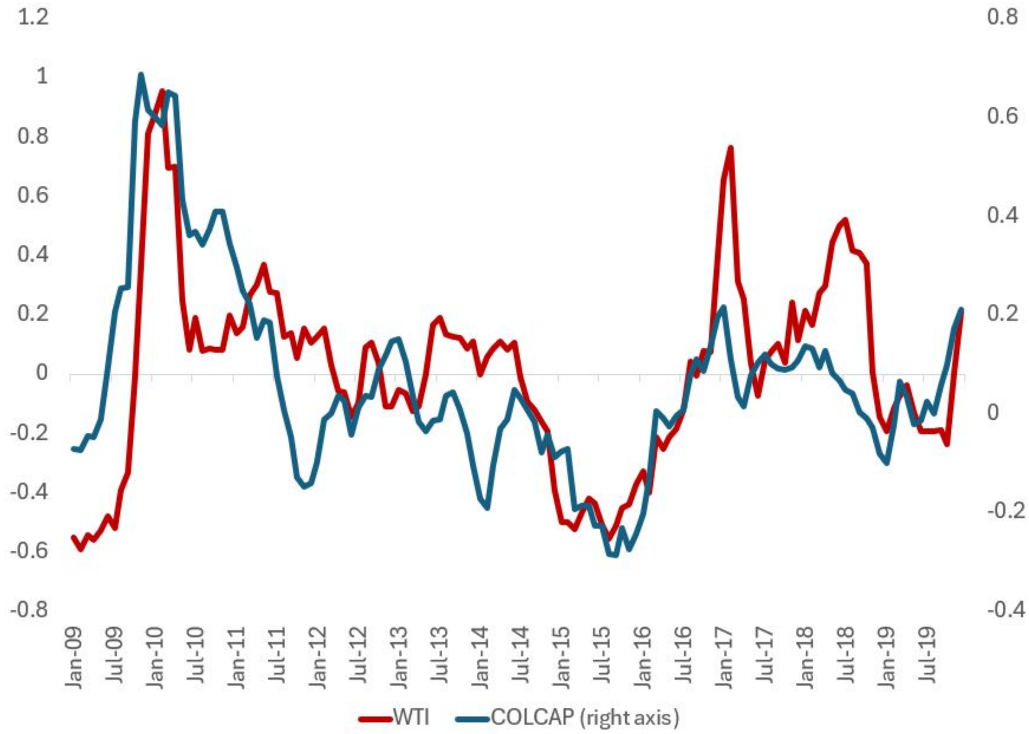



Figure 10: Annual growth of WTI and COLCAP
*Note:* The figure reports the annual percentage growth of the West Texas Intermediate oil price and the COLCAP index over the sample period. The COLCAP series is plotted on the right axis. The positive co-movement between the two series suggests that Colombian stock market valuations are strongly exposed to international oil price dynamics.

This interpretation is consistent with the results reported above. Unlike the BEI-based decomposition, which directly exploits revisions in inflation expectations, the COLCAP-based decomposition relies on a financial variable that is affected by multiple forces, including discount rates, expected profits, risk premia, external financial conditions, and commodity prices. The strong association between the COLCAP and WTI suggests that the stock market signal is not clean enough to isolate the inflation-expectations channel emphasized in this paper. Hence, the fact that the price puzzle appears mainly in the monetary policy component when using COLCAP should not be interpreted as evidence against the central bank information channel. Rather, it suggests that, in the Colombian context, the stock market response around policy announcements may be too strongly influenced by oil price movements to provide a reliable measure of central bank information.

# Concluding Comments

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# Appendices

## A Orthogonalization of surprises

Following Jarociński and Karadi (2020), let

$$M_t = \begin{bmatrix} MPS_t & \Delta BEI_t \end{bmatrix}$$

denote the vector of high-frequency surprises, where $MPS_t$ is the monetary policy surprise and $\Delta BEI_t$ is the change in break-even inflation around the announcement. We assume that, within the announcement window, these surprises are driven only by two structural shocks, so that

$$M_t = U_t C, \qquad U_t = \begin{bmatrix} u_t^m & u_t^i \end{bmatrix},$$

where $u_t^m$ denotes the conventional monetary policy shock and $u_t^i$ denotes the central bank information shock. The matrix $C$ collects the contemporaneous effects of these shocks on the observed high-frequency surprises.

To construct the orthogonal components, the procedure first apply a QR decomposition to the matrix of observed surprises,

$$M = QR,$$

where $Q'Q = I$. The QR decomposition provides an orthogonal basis for the two-dimensional space spanned by the observed surprises. However, these orthogonal components do not yet have an economic interpretation. Therefore, the method rotate this basis using an orthogonal rotation matrix

$$P(a) = \begin{bmatrix} \cos(a) & \sin(a) \\ -\sin(a) & \cos(a) \end{bmatrix},$$

where the angle $a$ is chosen so that the resulting impact matrix satisfies the required sign restrictions. In particular, after the rotation we obtain

$$U = QP(a)D, \qquad C = D^{-1}P(a)'R,$$

where $D$ is a diagonal scaling matrix. This normalization implies that the monetary policy surprise can be additively decomposed as

$$MPS_t = u_t^M + u_t^I.$$

with the identifying sign restrictions

$$c_M < 0, \qquad c_I > 0.$$

Thus, a conventional monetary policy shock is identified as an unexpected tightening that raises the monetary policy surprise and reduces break-even inflation, while a central bank information shock is identified as an unexpected tightening associated with an increase in break-even inflation. The QR decomposition is therefore used only to extract orthogonal components from the observed surprise space; the economic interpretation is obtained after rotating those components so that the sign restrictions are satisfied.